

Notes: Bordalo, Coffman, Gennaioli, and Shleifer (QJE, 2016)

Stereotypes

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1 Big picture

- **Question.** How do people form stereotypes about groups, and why do stereotypes often contain a “kernel of truth” while still producing biased beliefs?
- **Core idea.** The paper models stereotypes using Kahneman and Tversky’s representativeness heuristic. A type is stereotypical of group G if it is more frequent in G than in a relevant comparison group $\bar{G}$.
- **Key formula.** The representative type is not necessarily the most common type in G . It is the type with the highest likelihood ratio:

$$R(t, G, \bar{G}) = \frac{\Pr(t \mid G)}{\Pr(t \mid \bar{G})}.$$

- **Why stereotypes can be wrong.** Because the mind focuses on relative differences, rare but diagnostic types can be overweighted. A 65+ person is representative of Florida relative to the United States even though most Florida residents are not elderly.
- **Kernel of truth.** Stereotypes are rooted in real distributional differences across groups. If men are overrepresented in the right tail of math scores, the stereotype that men are better at math has a true diagnostic source, but the stereotype can exaggerate average gender differences.
- **Context dependence.** Stereotypes depend on the comparison group. The stereotype of the Irish may be red hair relative to Europeans, but Catholic identity relative to Scots.
- **Main theoretical result.** When group distributions differ by means, stereotypical beliefs exaggerate mean differences. When they differ by variances, stereotypical beliefs exaggerate variance differences.
- **Empirical strategy.** The paper tests the theory in two ways: a controlled lab experiment that changes the comparison group while holding the target group fixed, and survey evidence on beliefs about U.S. political groups.
- **Main empirical result.** Beliefs are distorted toward representative types. In political data, perceived ideological gaps are larger than true gaps for 109 of 111 issue-year observations.
- **Economic takeaway.** Stereotypes are neither fully rational statistical discrimination nor purely false prejudice. They are simplified mental representations that overweight what is distinctive.

2 Incremental contribution of the paper

- **From rational statistical discrimination to distorted belief formation.** Earlier economic models often treat stereotypes as rational beliefs about group averages. This paper shows how beliefs can be grounded in true group differences but still systematically distorted.
- **A formal model of the social-cognition view.** Social psychology emphasizes that stereotypes highlight distinctive group features. The paper gives that idea a precise economic model based on likelihood ratios and distorted probability weights.

- **A unifying role for representativeness.** The paper connects social stereotypes, base-rate neglect, diagnostic overreaction, and extrapolation-style prediction errors through a common mechanism: representative types come to mind first.
- **Context dependence as a sharp prediction.** The paper’s most distinctive theoretical prediction is that beliefs about a fixed target group change when the reference group changes. This is difficult to explain with standard rational-expectations models.
- **Direct experimental evidence.** The lab design holds the target group fixed but changes the comparison group. This directly identifies the causal role of representativeness in belief distortion.
- **Field evidence using political stereotypes.** The paper shows that beliefs about liberals and conservatives exaggerate true differences and that the magnitude of exaggeration is predicted by the representativeness of tail positions.
- **Beyond likelihood-based categorization.** In contrast to models in which people focus on likely or modal types, this paper shows that people often focus on diagnostic types, which can be unlikely in absolute terms.
- **Practical contribution.** The framework helps explain why small or tail-based differences can influence important economic decisions in labor markets, education, politics, finance, and social interaction.

3 Conceptual environment and measurement

3.1 Groups, types, and comparison groups

The decision maker needs to assess the distribution of types in a target group G . The set of possible types is T . Types can be unordered, such as hair colors or occupations, or ordered, such as income, ideology, age, or ability.

The model requires three objects:

- **Target group G .** The group being evaluated, such as Floridians, women, conservatives, or a sector of firms.
- **Type space T .** The dimension on which the group is evaluated, such as age brackets, math scores, or issue positions.
- **Reference group $\bar{G}$.** The comparison group relative to which distinctiveness is judged.

Interpretation: stereotypes are not formed in isolation. They are comparative objects. The same group can have different stereotypes in different contexts because $\bar{G}$ changes.

3.2 Representativeness

Representativeness is defined by a likelihood ratio:

$$R(t, G, \bar{G}) \equiv \frac{\pi_{t,G}}{\pi_{t,\bar{G}}} = \frac{\Pr(T = t \mid G)}{\Pr(T = t \mid \bar{G})}.$$

A type is representative when it is relatively more common in the target group than in the comparison group.

Important distinction:

- $\Pr(t \mid G)$ asks whether type t is common in group G .
- $\Pr(t \mid G) / \Pr(t \mid \bar{G})$ asks whether type t is diagnostic of group G relative to $\bar{G}$.

Interpretation: a type can be highly representative even if it is rare. This is why stereotypes can focus on tails of distributions.

3.3 Stereotypical beliefs

The decision maker stores the full distribution but retrieves a distorted version that overweights representative types. The stereotype distribution is:

$$\pi_{t,G}^{st} = \pi_{t,G} \frac{h_t(R(t, G, \bar{G}))}{\sum_{s \in T} \pi_{s,G} h_s(R(s, G, \bar{G}))}.$$

The weight $h_t(\cdot)$ is increasing in the representativeness of type t and decreasing in the representativeness of other types.

Interpretation: the distortion is not that people mechanically confuse $\Pr(t \mid G)$ with $\Pr(G \mid t)$. The distortion is attentional. Representative types are easier to recall and receive too much weight.

3.4 Functional forms

The paper discusses several possible weighting rules:

- **Rank-based truncation.** The decision maker recalls only the top d representative types and neglects the rest.
- **Rank-based discounting.** Less representative types are discounted smoothly according to their rank.
- **Representativeness-based discounting.** Weights increase continuously in the likelihood ratio.

Interpretation: the key results do not require a single exact weighting rule. They rely on the general property that more representative types receive more weight.

4 Models and theoretical results

4.1 Likely versus representative types

The modal type is the most likely type in the group. The exemplar is the most representative type:

$$t^* \in \arg \max_t \frac{\pi_{t,G}}{\pi_{t,\bar{G}}}.$$

These need not be the same. The Florida example shows why: the modal age group is 20–44, but the representative age group is 65+, because Florida’s elderly share is higher relative to the U.S. benchmark.

Interpretation: stereotype errors are especially large when the representative type is unlikely or extreme.

4.2 Proposition 1: When representative types are unlikely

If two groups have similar likelihood rankings, then the modal type cannot be the exemplar for at least one group. In plain language, when groups look broadly similar, their most distinctive differences often lie in relatively small deviations or tails.

If the two groups have opposite likelihood rankings, the modal type is the exemplar for each group. In this case, stereotypes are more likely to be accurate because the most diagnostic type is also common.

Interpretation: stereotypes are most misleading when groups are similar but differ in unlikely tails.

4.3 Proposition 2: Mean exaggeration

Suppose the likelihood ratio is monotonic in ordered type t . If high types are relatively more frequent in G , then:

$$E^{st}(t \mid G) > E(t \mid G) > E(t \mid \bar{G}) > E^{st}(t \mid \bar{G}).$$

The stereotype exaggerates a true mean difference. The group with the higher true mean is believed to have an even higher mean; the comparison group is believed to have an even lower mean.

Interpretation: this is the formal kernel-of-truth result. Stereotypes point in the right direction but overshoot.

4.4 Proposition 3: Variance exaggeration

If group G has more mass in both tails than $\bar{G}$, then the representative types of G are extreme types. The stereotype exaggerates the variance of G :

$$Var^{st}(t \mid G) > Var(t \mid G) > Var(t \mid \bar{G}) > Var^{st}(t \mid \bar{G}).$$

The means may remain correct if the distributions are symmetric, but perceived dispersion becomes too high for the group with more extreme types.

Interpretation: a group can be stereotyped as volatile, risky, bimodal, or extreme even when its average is not very different.

4.5 Examples and economic intuition

- **Florida residents.** The 65+ age bracket is representative of Florida relative to the United States, but it is not the most likely age bracket. This shows how a stereotype can have a real basis and still be inaccurate.

- **Gender and math.** Men and women have similar SAT math distributions, but men are overrepresented in the high right tail. Representativeness can lead people to exaggerate average male-female differences in math ability.
- **Medical tests and base-rate neglect.** A positive test result makes sickness diagnostic, so the sick type comes to mind too strongly. The decision maker moves in the correct direction but overreacts to diagnostic information.
- **Finance and hot sectors.** A hot sector brings successful firms to mind, leading investors to overweight success stories and underweight failure risk.

5 Empirical specifications and evidence

5.1 Laboratory test of context dependence

The experiment constructs artificial groups of ordinary objects. The target group G is held fixed, while the comparison group $\bar{G}$ changes. In the T-shirts experiment, girls are the target group and boys are the comparison group.

In the control condition, the two groups have nearly identical color distributions. In the representativeness condition, the comparison group is changed so that a certain color becomes distinctive for girls or boys.

Main prediction:

$$\text{same } G + \text{different } \bar{G} \Rightarrow \text{different beliefs about } G.$$

Key result: in the T-shirts design, participants in the representativeness condition are 10.5 percentage points more likely to report the less frequent but representative color as modal. In pooled unordered-type experiments, participants are 9.3 percentage points more likely to choose the less frequent type as modal when it is representative.

Interpretation: because the target group itself is unchanged, the belief shift is evidence that the comparison group causally affects recall and judgment.

5.2 Political stereotype data

The paper uses two data sets:

- **Graham, Nosek, and Haidt data.** Respondents report their own positions and beliefs about typical liberals and conservatives on 45 moral issues. Positions are measured on a 1–6 scale.
- **ANES data.** Respondents report their own positions and beliefs about Democratic and Republican party positions on 10 recurring political issues from 1964 to 2012. Positions are measured on a 1–7 scale.

The true distributions come from respondents’ own positions. Believed distributions are summarized by believed typical positions.

Interpretation: political beliefs provide a real-world setting where group stereotypes are economically and socially relevant, and where true and perceived group positions can be compared.

5.3 Kernel-of-truth regression

The first empirical specification tests whether beliefs about group G depend not only on the true mean of G but also on the true mean of the comparison group $\bar{G}$:

$$E^B(t \mid G) = \alpha + \beta E(t \mid G) + \gamma E(t \mid \bar{G}) + \varepsilon.$$

The model predicts $\beta > 0$ and $\gamma < 0$. A high comparison-group mean makes the target group look less extreme, while a low comparison-group mean makes it look more extreme.

Interpretation: the negative coefficient on the comparison group is direct evidence against a purely rational expectation based only on G 's own distribution.

5.4 Representativeness regression

The second empirical specification asks whether belief distortions are predicted by the representativeness of tail positions. Define the average representativeness of the conservative right tail as:

$$R_H^{cons} = \frac{\sum_{t \in H} \pi_{t,cons}}{\sum_{t \in H} \pi_{t,lib}}.$$

The paper estimates whether conservative beliefs rise and liberal beliefs fall as the conservative right tail becomes more representative:

$$E^{st}(t \mid cons) = E(t \mid cons) + \lambda_{cons}(R_H^{cons} - 1),$$

$$E^{st}(t \mid lib) = E(t \mid lib) - \lambda_{lib}(R_H^{cons} - 1).$$

Interpretation: the regressions test the mechanism, not merely the existence of exaggeration. They ask whether the amount of exaggeration is tied to diagnostic tail types.

6 Identification and empirical design

6.1 What the model can explain

The model explains stereotypes that are based on distinctive statistical features of groups. It is especially powerful when stereotypes are directionally related to real differences but too extreme.

It can explain:

- exaggeration of group differences;
- context-dependent beliefs;
- overweighting of rare but diagnostic types;
- base-rate neglect and overreaction to diagnostic signals;
- tail-driven stereotypes in education, politics, labor markets, and finance.

6.2 What the laboratory design identifies

The lab design directly identifies context dependence because the target group does not change across treatments. Only the comparison group changes.

This eliminates many confounds present in social settings, such as history, prejudice, media exposure, or actual changes in the target group.

Interpretation: if beliefs about the same target group shift when only the reference group changes, then stereotypes are comparative rather than absolute.

6.3 What the political data can and cannot identify

The political data do not randomly assign comparison groups. They show whether naturally occurring beliefs are consistent with the model.

The strongest political evidence is that:

- perceived differences are systematically larger than true differences;
- beliefs about one group depend negatively on the true position of the other group;
- tail representativeness predicts the direction and magnitude of belief distortion.

Interpretation: the political evidence supports the representativeness mechanism, but it is observational and cannot by itself prove causality.

6.4 Relation to alternative models

- **Statistical discrimination.** Standard statistical discrimination predicts accurate beliefs about group distributions. The paper rejects this as a complete description because beliefs exaggerate true gaps.
- **Pure prejudice.** Prejudice may matter in some settings, but the model explains why stereotypes can be flattering, changing, and partially accurate.
- **Likelihood-based categorization.** A model where people focus on the most likely types cannot easily explain why rare but diagnostic types dominate stereotypes.
- **Availability.** Availability can shape recall, but the political data show a distinct role for representativeness after controlling for likelihood in supplemental analysis.

7 Tables and figures

7.1 Florida age table and Figure I: examples of tail-driven stereotypes

Read:

- The Florida age table shows that Florida and U.S. age distributions are very similar.
- The 65+ group is only 17.4% of Florida residents, while the 20–44 group is 31.7%.

- Yet the 65+ group is representative of Florida because the elderly share is higher in Florida than in the United States.
- Figure I shows SAT math distributions by gender. The male and female distributions are similar, but men are overrepresented in the right tail.
- These two examples show the same mechanism: stereotypes often arise from relative overrepresentation in a tail, not from the most likely group type.

Economic message: The stereotype has a kernel of truth because it is based on a real relative difference. It becomes misleading because people overweight the diagnostic type and underweight the rest of the distribution.

<i>age</i>	0 – 19	20 – 44	45 – 64	65+
Florida	24.0%	31.7%	27.0%	17.4%
US	26.9%	33.6%	26.4%	13.1%

(a) Florida age-distribution example

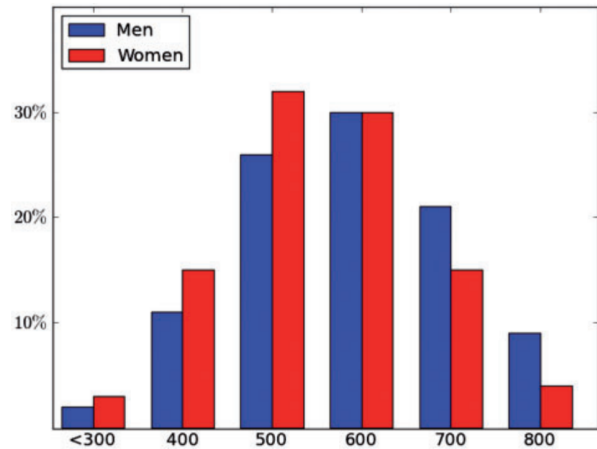


FIGURE I
SAT Mathematics Scores by Gender (2013)

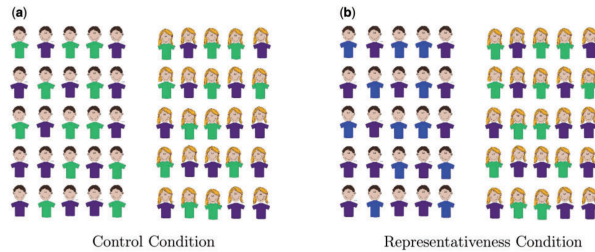
(b) Figure I: SAT mathematics scores by gender

7.2 Figure II and Figure III, Example 1: context dependence and political distributions

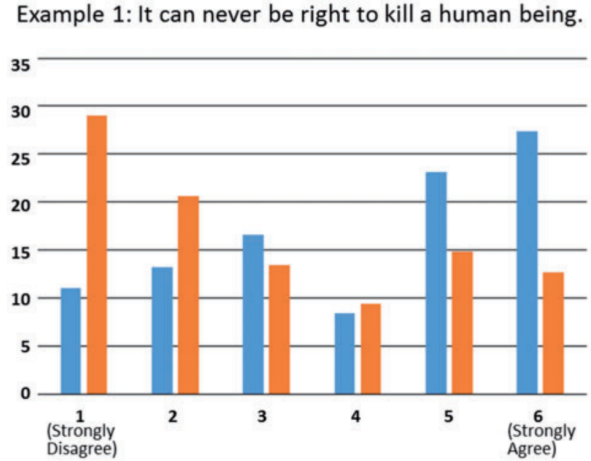
Read:

- Figure II is the main experimental illustration. In the control condition, girls and boys have nearly identical color distributions. In the representativeness condition, the boys' distribution changes, making green distinctive for girls and blue distinctive for boys.
- The target group is unchanged, but the stereotype changes because the comparison group changes.
- Figure III, Example 1 shows a political issue where the modal and representative conservative positions coincide.
- In this case, stereotyping creates less distortion because the representative type is also common.

Economic message: The T-shirts experiment isolates the mechanism: representativeness changes beliefs even when the target distribution is held fixed. Example 1 shows why distortion is smaller when representative and modal types coincide.



(a) Figure II: T-shirts experiment



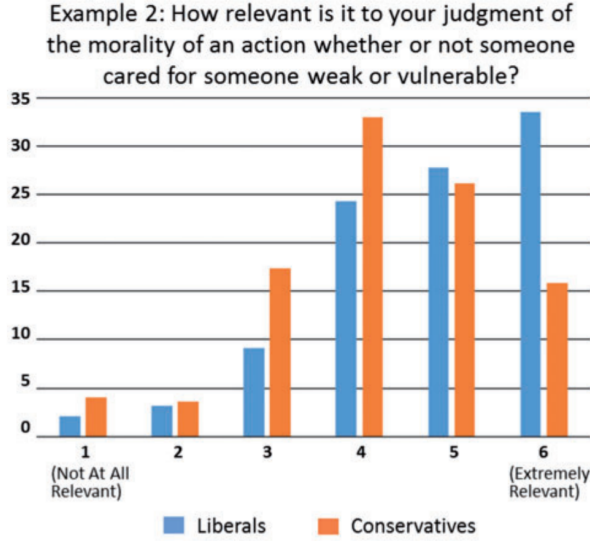
(b) Figure III, Example 1

7.3 Figure III, Example 2 and Figure IV: exaggeration of political differences

Read:

- Figure III, Example 2 shows a political issue where representative conservative positions are not the modal conservative positions.
- This generates larger belief distortion because the representative types are away from the center of the true distribution.
- Figure IV compares believed and true ideological differences across GNH and ANES observations.
- Almost all observations lie above the 45-degree line: believed conservative-liberal gaps exceed true gaps.
- The paper reports exaggeration in 109 of 111 observations, with average exaggeration of about 0.62 scale positions.

Economic message: Political stereotypes are not random errors. They systematically exaggerate differences in the direction of representative positions.



(a) Figure III, Example 2

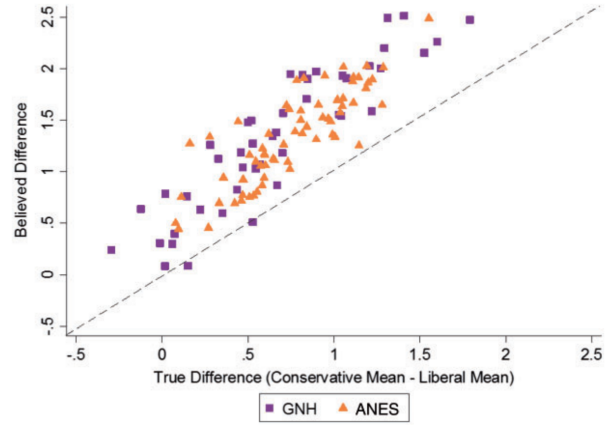


FIGURE IV
Exaggeration of Differences

(b) Figure IV: exaggeration of differences

7.4 Tables I and II: regression evidence on context and representativeness

Read:

- Table I tests whether information about $\bar{G}$ predicts beliefs about G .
- For beliefs about conservatives, the true conservative mean is positive and the true liberal mean is negative. For beliefs about liberals, the pattern reverses.
- This is the regression version of context dependence: beliefs about one group depend on the other group's distribution.
- Table II tests whether average representativeness of conservative tail types predicts beliefs.
- The tail-representativeness coefficient is positive for beliefs about conservatives and negative for beliefs about liberals, matching the model.

Economic message: The regression evidence supports both the kernel-of-truth logic and the more specific representativeness mechanism. Beliefs exaggerate real differences, and exaggeration is strongest when tail types are more diagnostic.

TABLE I
INFORMATION ABOUT $-G$ PREDICTS BELIEFS ABOUT G

OLS predicting believed mean of group G for each issue						
	$G = \text{conservatives}$			$G = \text{liberals}$		
	GNH	ANES	Pooled	GNH	ANES	Pooled
True mean position of conservatives for issue	1.02**** (0.097)	0.98**** (0.133)	0.96**** (0.076)	-0.21**** (0.060)	-0.19 (0.116)	-0.25**** (0.060)
True mean position of liberals for issue	-0.35**** (0.106)	-0.86**** (0.134)	-0.58**** (0.131)	0.987**** (0.066)	0.39**** (0.106)	0.73**** (0.135)
Constant	1.51**** (0.195)	3.35**** (0.269)	2.35**** (0.279)	0.69**** (0.122)	2.58**** (0.249)	1.56**** (0.270)
R^2	0.83	0.53	0.66	0.92	0.32	0.68
Obs. (clusters)	45 (45)	66 (10)	111 (55)	45 (45)	66 (10)	111 (55)

Notes. Standard errors in parentheses, clustered at the issue level. *, **, ***, and **** denote significance at the 10%, 5%, 1%, and 0.1% levels, respectively. In pooled specifications, we include a dummy variable indicating whether the observation came from ANES data set.

(a) Table I: information about $\bar{G}$ predicts beliefs about G

TABLE II
AVERAGE REPRESENTATIVENESS OF TAIL POSITIONS PREDICTS BELIEFS

OLS Predicting Believed Mean of Group G for Each Issue						
	$G = \text{conservatives}$			$G = \text{liberals}$		
	GNH	ANES	Pooled	GNH	ANES	Pooled
True mean position of group G for issue	0.78**** (0.06)	0.24** (0.08)	0.51**** (0.09)	0.72**** (0.05)	0.18**** (0.05)	0.41**** (0.10)
Avg. rep. of tail types for conservatives, R_H^{cons}	0.19** (0.07)	0.55** (0.22)	0.25*** (0.08)	-0.14** (0.06)	-0.12* (0.07)	-0.24**** (0.05)
Constant	1.01**** (0.26)	2.60**** (0.45)	1.84**** (0.29)	0.93**** (0.22)	2.71**** (0.25)	2.01**** (0.32)
R^2	0.82	0.48	0.60	0.91	0.31	0.70
Obs. (clusters)	45 (45)	66 (10)	111 (55)	45 (45)	66 (10)	111 (55)

Notes. Standard errors in parentheses, clustered at the issue level. *, **, ***, and **** denote significance at the 10%, 5%, 1%, and 0.1% levels, respectively. In pooled specifications, we include a dummy variable indicating whether the observation came from ANES data set.

(b) Table II: tail representativeness predicts beliefs

8 Short summary

- The paper models stereotypes as distorted probability assessments based on representativeness.
- A type is representative when it is relatively more frequent in a target group than in a comparison group.
- Stereotypes therefore depend on the reference group, not just on the target group.
- Stereotypes can contain a kernel of truth because they are based on real differences across groups.
- They can still be biased because they overweight diagnostic types and underweight nonrepresentative types.
- Theoretical results show that stereotypes exaggerate mean differences under monotone likelihood ratios and exaggerate variance differences when a group is overrepresented in the tails.
- The lab experiment shows context dependence directly: changing the comparison group changes beliefs about an unchanged target group.
- Political survey data show that perceived ideological differences exceed true differences and that tail representativeness predicts belief distortion.
- The model provides a unified explanation for social stereotypes, base-rate neglect, diagnostic overreaction, and tail-driven belief errors.

9 Conclusion

9.1 Core conclusion

Bordalo, Coffman, Gennaioli, and Shleifer develop a parsimonious model in which stereotypes arise because decision makers overweight the most representative types of a group. The representative type is the type most diagnostic of the group relative to a comparison group. This mechanism explains why stereotypes often reflect real differences but exaggerate them.

9.2 Economic intuition

The economic intuition is that people compress a complex distribution into a simplified mental representation. The compression is not based on what is most common. It is based on what stands out. Because what stands out depends on the comparison group, stereotypes are context dependent. Because what stands out can be rare or extreme, stereotypes can be inaccurate.

9.3 Incremental contribution revisited

The paper advances the economics of beliefs by moving beyond two polar views: stereotypes as rational statistical discrimination and stereotypes as purely false prejudice. Its contribution is to show how a psychologically realistic recall rule produces beliefs that are partly grounded in facts but systematically biased. It also connects stereotypes to a broader family of belief distortions driven by representativeness.

9.4 Implications

The model implies that small distributional differences can have large perceived effects if they occur in representative types. This has implications for discrimination, political polarization, educational choices, hiring, investing, and any setting where people make decisions about individuals based on group membership.

For policy and institutional design, the model suggests that merely providing average information may not be enough. If people continue to focus on distinctive tails, debiasing may require presenting full distributions or directly highlighting base rates and within-group heterogeneity.

9.5 Limitations

The model focuses on representativeness and does not capture every source of stereotyping. Availability, media exposure, motivated reasoning, prejudice, and strategic communication may also shape beliefs. The political evidence is observational, so it supports the model's mechanism but does not by itself identify causality. The lab evidence is causal but uses simplified artificial groups.

9.6 Takeaway

The enduring takeaway is that stereotypes are simplified representations of group distributions that emphasize what is distinctive. This makes stereotypes cognitively useful but dangerous: they can identify real differences while exaggerating them and overlooking the majority of cases within each group.